

Injection Techniques — Complete Route Reference

Transdermal, intradermal, subcutaneous, intramuscular, intravenous — comparison, needle selection, depth, and technique

CLINICAL REFERENCE

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THE 5 INJECTION ROUTES — COMPLETE COMPARISON

Route	Layer Targeted	Depth	Needle Gauge	Angle	Onset	Common Uses in Med Spa
Transdermal (topical)	Stratum corneum → epidermis (passive diffusion)	Surface only — no needle	No needle — or microneedle 0.25–0.5mm	N/A	Slow — hours to days	Topical anesthetics (EMLA, LMX), topical retinoids, microneedling delivery enhancement
Intradermal (ID)	Dermis — within the skin layer itself	1–2mm	26–30g, 3/8" to 1/2"	10–15°	Slow — localized	Allergy testing, Botox superficial (select zones), Skinvive/HA microdroplets, local anesthetic wheal
Subcutaneous (SubQ)	Hypodermis — fatty tissue below dermis	4–10mm (varies by fat depth)	25–31g, 5/8" (standard)	45° (pinch skin) or 90° (obese patient)	Slow to moderate — 15–30 min	Testosterone cypionate (women), semaglutide, tirzepatide, peptide injections, vitamin injections, insulin
Intramuscular (IM)	Muscle belly — beneath fascia	25–38mm depending on site and body habitus	21–25g, 1" to 1.5"	90°	Faster than SubQ — 10–15 min	B12, B-complex, MIC/Lipo, glutathione, biotin, vitamin D, testosterone cypionate (men), GLP-1 (some protocols), vaccines
Intravenous (IV)	Directly into bloodstream via vein	Into lumen of vein	18–22g angiocatheter	15–30°	Immediate — seconds to minutes	IV hydration, NAD+, glutathione push, vitamin C, Myers cocktail, all IV therapy drips

Intradermal (ID)

10–15° · 26–30g · 1–2mm

Technique

Position: Bevel UP — always

Angle: 10–15° nearly parallel to skin

Insertion: Advance just under skin surface — 2–3mm

Endpoint: Visible wheal (blister-like bump) confirms correct placement

No aspiration — intradermal layer has no vessels of concern

No massage after — disrupts wheal

Med Spa Applications

Skinvive/HA microdroplets — cheek intradermal grid

Botox — periorbital lower lid (advanced)

Local anesthetic wheal before cannula entry

Allergy skin testing (not typically in med spa)

Volume per deposit: 0.01–0.05mL maximum

Common error: Too steep angle = SubQ — no wheal forms

Subcutaneous (SubQ)

45–90° · 25–31g · 4–10mm

Technique

Pinch technique: Grasp subcutaneous fat between thumb and finger — lifts away from muscle

Angle: 45° for lean patients; 90° for patients with more subcutaneous fat

Insertion: Quick, smooth — insert to hub or appropriate depth

Aspiration: Not required for SubQ (vessels rare) — but standard for testosterone

Inject slowly: Slow pressure to minimize discomfort and site reaction

Release pinch before withdrawing to prevent tracking

Med Spa Applications

GLP-1 (semaglutide, tirzepatide) — weekly injection

Peptide injections (BPC-157, CJC-1295, TB-500, etc.)

Testosterone cypionate — women (low dose)

Vitamin/mineral injections (alternative to IM)

Best sites: Abdomen (2" from navel), thigh, upper arm

Rotate sites: Lipohypertrophy risk with repeated same-site injection

Intramuscular (IM)

90° · 21–25g · 25–38mm

Technique

Angle: 90° — straight into muscle

Spread skin: Z-track technique or spread skin taut (not pinch)

Med Spa Applications

B12, B-complex — deltoid or ventrogluteal

Intravenous (IV)

15–30° · 18–22g · Into vein lumen

Peripheral IV Technique

Apply tourniquet 3–4 inches above site — dilates vein

Med Spa IV Applications

IV hydration (NS, LR) — rapid onset

NAD+ infusion — 1–4 hours depending on dose

Aspirate: Pull back plunger 5–10 seconds before injecting — blood = wrong placement, reposition Inject rate: 1mL per 10 seconds. Do not inject faster. Withdraw smoothly at same angle as insertion Apply pressure 30 seconds post-injection. Light massage (unless Botox/specific product)	MIC/Lipo-C injections — deltoid preferred Glutathione IM — deltoid Testosterone cypionate — men (ventrogluteal preferred) Vitamin D (high dose) — deltoid or ventrogluteal Biotin — deltoid	Palpate vein — feel for bounce and track. Do not insert blind. Angle: 15–30°, bevel up, pierce skin then advance at shallower angle into vein Flashback: Blood return in catheter chamber confirms vein entry Advance catheter off needle, withdraw needle, release tourniquet Flush with 10mL NS before connecting drip — confirm patency	Myers cocktail (Mg, B-complex, Vit C, B12) High-dose vitamin C — immune support Glutathione IV push — brightening, antioxidant Must assess: BP and pulse before every IV infusion
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IM INJECTION SITES – LANDMARKS, VOLUME LIMITS & WHEN TO USE EACH

Site	Landmark	Max Volume	Preferred For	Avoid If	Patient Position
Deltoid (upper arm)	2–3 finger widths below acromion process, lateral aspect of arm	1–2mL	B12, MIC, glutathione, biotin, vitamin D, most vitamin injections	Atrophied deltoid, lymphedema of arm, axillary node dissection same side	Seated, arm relaxed at side or slightly abducted
Ventrogluteal	Place heel on greater trochanter, index finger on ASIS, middle finger toward iliac crest — inject in the V between fingers	3–5mL	Testosterone cypionate (men), large volume medications, hormones	Insufficient muscle mass, unable to identify landmarks	Side-lying preferred; standing with weight on opposite leg acceptable
Dorsogluteal	Upper outer quadrant of buttock — above and lateral to midpoint	3–5mL	Historically used but largely replaced by ventrogluteal	Sciatic nerve proximity risk — NOT recommended as first choice	Prone or side-lying
Vastus lateralis (thigh)	Outer middle third of thigh — anterolateral surface	2–5mL	Self-injection, pediatric, poor deltoid or gluteal access	Poor muscle mass, circulatory compromise in leg	Seated or supine with knee slightly flexed

NEEDLE SELECTION QUICK REFERENCE

Route	Gauge	Length	Viscosity Fit	Notes
Intradermal	26–30g	3/8"–1/2"	Very thin solutions	Bevel up always. Short bevel preferred for ID — long bevel increases SubQ risk.
SubQ — peptides / GLP-1	29–31g	4mm–8mm (5/16")	Thin solutions	Insulin-type pen needles for GLP-1. Short fine needles reduce pain significantly.
SubQ — testosterone (women)	25–27g	5/8"	Oil-based (moderate viscosity)	Testosterone is oil-based — cannot use 29–31g. Requires at minimum 25g.
IM — vitamin injections	23–25g	1"	Aqueous solutions	25g reduces discomfort for thin aqueous solutions. 23g if faster injection needed.
IM — testosterone (men)	21–23g	1"–1.5"	Oil-based (higher viscosity)	Oil-based testosterone requires larger gauge. 1.5" for patients with higher BMI to reach muscle.
IV — angiocatheter	18–22g	1"–1.25" catheter	All IV fluids	18–20g for viscous infusions (high-dose vitamin C). 20–22g for standard IV hydration and vitamin drips.

Z-TRACK TECHNIQUE – WHEN & WHY

What it is	Pull skin 2–3cm laterally before injection. Release after withdrawing. Creates a zigzag path that seals medication in muscle and prevents tracking to surface.
When to use	Iron dextran (mandatory), any medication that stains skin or is highly irritating, oil-based

INJECTION COMPLICATIONS – RECOGNITION & RESPONSE

Blood in aspirate	Withdraw needle. Do not inject. Apply pressure. Prepare new dose in new syringe. Select alternate site.
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	hormones (testosterone), any IM injection where leakage is a concern	Hematoma	Apply firm pressure 2–5 minutes. Elevate if limb. Document. Monitor for expansion. Ice after 24 hours.
Technique	1) Clean site. 2) Pull skin lateral with non-dominant hand. 3) Insert needle at 90°. 4) Aspirate. 5) Inject slowly. 6) Wait 10 seconds. 7) Withdraw needle. 8) Release skin. 9) Apply pressure — do NOT massage (disturbs seal).	Abscess / infection	Warmth, redness, swelling days after injection. Antibiotics. Incision and drainage if fluctuant. Review sterile technique.
Never massage after Z-track	Massaging breaks the Z-track seal — allows medication to track back to SubQ layer or skin surface	Lipohypertrophy	Fatty lump from repeated same-site SubQ injection. Prevention: site rotation every injection. No treatment once present except site avoidance.
		Nerve strike (IM)	Patient reports sharp shooting pain or numbness during IM injection. Withdraw immediately. Do not inject. Reassess landmark. Most resolve fully.
		Anaphylaxis	Urticaria + hypotension + bronchospasm = anaphylaxis. Epinephrine 0.3–0.5mg IM (thigh) IMMEDIATELY. Call 911. Lay patient supine, legs elevated. Have protocol posted and epi auto-injector in every room.

ASPIRATION — WHEN REQUIRED, WHEN OPTIONAL, WHEN NOT NEEDED

Route	Aspiration	Rationale
Intradermal	Not required	Dermal layer has no large vessels. Wheal formation confirms correct placement.
Subcutaneous	Not required (current CDC/WHO guidance)	SubQ layer has minimal vascular risk for standard medications. Exception: some providers still aspirate for testosterone SubQ.
Intramuscular — deltoid, thigh	Not required per current CDC guidance for vaccines	Low vessel density. However — many med spa providers aspirate for all IM injections as a safety standard. Either approach is defensible.
Intramuscular — ventrogluteal/dorsogluteal	Aspirate — standard practice	Closer proximity to superior gluteal artery. Aspirate 5–10 seconds before injecting. Blood = reposition.
Intravenous	Flashback confirms placement	Blood return in catheter confirms vein entry. Flush with NS to confirm patency before connecting drip.

- ★ **One needle, one patient, one time.** Never recap a used needle. Never reuse a syringe. Never use a single-use vial for multiple patients.
- ★ **Sharps in sharps containers immediately** after withdrawal. Do not set on counter, pass to another person, or carry across the room.
- ★ **Room temperature medications inject more comfortably** — oil-based testosterone especially. Remove from refrigerator 30 minutes prior.
- ★ **Slow injection = less pain.** 1mL per 10 seconds for IM. Pressure pain is caused by rapid bolus delivery — not the needle. Train every provider on injection speed.
- ★ **Epinephrine must be in every injection room.** Every room. Every day. Every patient. Anaphylaxis is rare — but when it happens, seconds matter.